Indicator Functions

(with random variables)

Let X be a random variable.

Define

$$I_{\{X \geq 3\}} = \begin{cases} 1, & \text{if } X \geq 3 \\ 0, & \text{if } X < 3 \end{cases}$$

Then $I_{\{X \geq 3\}}$ is a discrete random variable. (Even if X is continuous!)

$I_{\{X \geq 3\}}$ takes on values 0 or 1.

It's expected value is

$$E[I_{\{X \geq 3\}}] = 0 \cdot P(I_{\{X \geq 3\}} = 0) + 1 \cdot P(I_{\{X \geq 3\}} = 1)$$

$$= P(I_{\{X \geq 3\}} = 1) = P(X \geq 3)$$

⊛ The expected value of an indicator is always the probability of the event being indicated!

Alternative Characterization of Recurrent and Transient States

Theorem: State i is recurrent if and only if

$$\sum_{n=1}^{\infty} P_{ii}^{(n)} = \infty.$$

Equivalently, i is transient if and only if

$$\sum_{n=1}^{\infty} P_{ii}^{(n)} < \infty.$$

<u>Proof</u>:

- Let i be transient and let $X_0 = i$.

- Let $N = \#$ returns to i at times ≥ 1.

- i is transient $\iff g_i < 1 \iff \underbrace{\frac{g_i}{1-g_i}}_{E[N \mid X_0 = i]} < \infty$

- $\infty > E[N \mid X_0 = i] = E\left[\sum_{n=1}^{\infty} I_{\{X_n = i\}} \,\middle|\, X_0 = i\right]$

- $$\infty > \sum_{n=1}^{\infty} E[I_{\{X_n = i\}} \mid X_0 = i]$$

$$= \sum_{n=1}^{\infty} P(X_n = i \mid X_0 = i)$$

$$= \sum_{n=1}^{\infty} P_{ii}^{(n)} \quad //$$